

WHAT HAPPENS AFTER SUCCESS: THE PERILS OF EXCELLENCE*

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ABSTRACT

This article examines how past performance influences the way an organization evolves, makes decisions, and adapts to its environment. It argues that compared to other periods of history, those that follow a lengthy interval of success will reveal companies that are especially apt to: (1) exhibit *inertia* in many aspects of structure and strategy-making process; (2) pursue *immoderation*, that is, adopt extreme process orientations; (3) manifest *inattention*, that is, reduce intelligence gathering and information processing activity; and (4) demonstrate *insularity* by failing to adapt to changes in the environment. An empirical analysis of the long-term histories of 36 companies provides tentative support for these notions.

INTRODUCTION

Administrative theorists are beginning to pay more attention to the way organizations evolve. But there remains much confusion about the factors that determine such change. This article will look into the role that past performance plays in corporate evolution. It will argue that enduring prior success will contribute very much to a company's attachment to past practices, to the evolution of its structure and strategy-making processes, and to its responsiveness to the environment.

Past performance may influence organizational behaviour in part by its impact on managerial learning. Performance, as we shall see, can influence search, scanning, experimentation (Lant and Montgomery, 1987; March, 1991), and hence organizational adaptation (Levinthal, 1991). But performance can also influence corporate evolution through its impact on organizational power and politics (Miller, 1993b), slack resources (Milliken and Lant, 1991), managerial confidence and motivation (Miller, 1990), and the homogeneity and intolerance of organizational cultures (Miller, 1993a). In short, we shall identify many paths through which performance can have an impact on subsequent organizational evolution.

And we expect there to be many effects from long-term success (Miller, 1990; 1993a; Milliken and Lant, 1991). Indeed, the periods of organizational history that follow long intervals of success are anticipated to be very different from those that succeed poor or modest performance. Our major hypotheses are that the former periods will be more apt to show companies exhibiting (1) *inertia* as

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they adhere more closely to the orientations of the past; (2) *immoderation* as strategy-making processes are taken to extremes; (3) *inattention*, that is, a reduction in information processing; and (4) *insularity* in ceasing to adapt their structure and processes to the changes in their environments.

These four alliterative hypotheses do not contrast organizations but rather their periods of organizational history. Entire histories are obviously too complex to be compared without first breaking them into thematic intervals. These can then be contrasted longitudinally or cross-sectionally with various purposes in mind. The comparison here will be between two kinds of periods: post-success or 'star' periods which follow at least three years of both superior profitability and growth *vis-à-vis* competitors; and post-mediocrity periods that follow poor-to-average relative performance. The hypotheses do not concern the effects of brief bouts of success but rather those of long intervals of superior performance.

ORGANIZATIONAL INERTIA

Hinings and Greenwood (1988), Miller (1990, 1993a), Miller and Friesen (1984) and Tushman and Romanelli (1985) argued that success induces organizations to converge on and adhere to established themes and practices. They refine existing strategies and structures and normally depart from the *status quo* only under the duress of crisis. Managerial learning and attributions, self-perpetuating power structure, and the effects of slack resources all help to explain how success gives rise to inertia.

Superstitious Reinforcement

Reinforcement models of learning suggest that successful behaviour will be repeated (Levitt and March, 1988; March, 1991). In an organizational context, managers tend to attribute success to some pet policy, strategy or process, which, as a result, becomes resistant to change. But given the ambiguity of cause-effect relationships in organizations, this attribution might be based as much upon superstition or error as upon accurate deduction (Lant and Montgomery, 1987; Milliken and Lant, 1991; Nystrom and Starbuck, 1984). In these cloudy and complex contexts, such attributions are hard to challenge. And so are the policies and practices based on them.

Attribution, Rationalization and Commitment

The leaders of thriving enterprises usually attribute their successes to the strategies that they themselves have developed (Staw and Fox, 1974). They are averse to abandoning these cherished and politically useful orientations. It is not surprising then that managers blame problems on uncontrollable and temporary external factors (Miller and Ross, 1975; Staw et al., 1983).

Moreover, when policies or strategies have produced consistent or exceptional rewards, managers are especially eager to take credit for and to openly commit to them (Bettman and Weitz, 1983; Staw et al., 1983). But such public commitment makes administrators reluctant to change since they want to appear consistent, and dislike being accused of 'backtracking' (Salancik and Meindl, 1984; Staw and Ross, 1978).

In short, corporate success encourages managers both to credit and take credit for their own policies, then to publicly endorse and tie their reputations to those policies, and ultimately to become bound to the preservation and extension of the policies. In contrast, in troubled companies there are likely to be fewer managers who are so enamoured of and tied to the ways of the past (Kiesler and Sproull, 1982; Lant et al., 1992).

Leadership: Self-Perpetuating Power, Institutional Legitimacy and Succession

Pfeffer (1981, p. 289) notes that 'power is stable in most organizations most of the time'. Top executives are hard to challenge because 'any act performed by the occupant of an office is seen as highly objectified and exterior' (Zucker, 1977; p. 729). The activities of high ranking executives are accorded legitimacy and often go largely unquestioned simply because of the power of the office. Moreover, many leaders, especially those in rich organizations, have the power and resources to keep themselves at the top (Pfeffer, 1981). Because executives' egos and reputations are so intimately tied to established structures and practices (Staw and Fox, 1977), organizational inertia can be very much amplified by a stable power distribution.

Clearly, the legitimacy that leaders and their policies enjoy among followers will be highest when they have garnered success in the marketplace. Even the power and resources available to top managers to perpetuate their tenure will be greatest then (Meyer and Zucker, 1989). In contrast, poor performance is more likely to produce the resource shortages and elicit the challenges to leadership that pave the way for change (Lant et al., 1992; McEachern, 1975).

If self-perpetuating power is a force for stability, then top management succession can sometimes induce change (Helmich, 1978; Miller, 1993b; Tushman et al., 1989). Although poor performance hastens the departure of the CEO, good performance can actually prolong a leaders' tenure. And it has been suggested that a new leader will have more of an impact on the practices of sick versus healthy organizations (Guest, 1962; Hambrick and Finkelstein, 1987; Tushman and Keck, 1990; Tushman et al., 1989). Periods of success, then, are the least likely to promote the leadership turnover that facilitates change, whereas troubled intervals are more apt to occasion invigorating transfers of power.

Slack Resources

Success allows organizations to accumulate slack resources – financial and reputational capital that they can use to avoid adapting to their environments (Cyert and March, 1963; Meyer, 1982). According to Milliken and Lant (1991, p. 138), '... slack acts as a buffer between the organization and environmental variation, reducing managers' perceived need to change. Because successful performance also increases the tendency to perceive environmental changes as temporary or unimportant, managers of successful organizations may use their slack resources to wait out the unfavourable environmental activity'.

From the arguments presented above, we derive our first hypothesis:

H1: Firms will display greater adherence to the orientations of the past after enduring success than after mediocre performance.

IMMODERATION: THE DEVELOPMENT OF EXTREMES

Success can evoke the enthusiasm, intolerance, and overwhelming consensus that leads managers to undertake more extreme modes of strategy-making. This will be especially evident in the way a firm interacts with its competitors and customers, in how much it innovates, in the risks it takes, and in whether it leads or follows its rivals.

Enthusiasm and Intolerance

Successful companies are said to have strong cultures (Deal and Kennedy, 1982; Peters and Waterman, 1982). They develop an enduring and pervasive set of shared beliefs and values which 'legitimate and institutionalize their present practices, structures, and influence distributions' (Pfeffer, 1981, p. 299). But it may be that good performance *creates* a cohesive and enthusiastic culture as much as the other way around. Policies believed to be sources of prestige or wealth elicit so much conviction that they are amplified and extended to extremes: for example, venturesome firms take even more risks while cautious ones become still more conservative; innovators pursue more grandiose inventions whereas tradition-bound firms come to eschew novelty entirely (Miller, 1990, 1993a).

Dissenters can act as a moderating influence in many organizations. But in long-successful companies, rebels are rarely tolerated because the policies they threaten are both sacred and highly rewarding to those in power. At the once-thriving General Motors, for example, many excellent suggestions regarding small cars, pollution controls and quality improvements were rejected out of hand because they challenged cherished beliefs. They were also sponsored by relatively disenfranchised groups such as engineers, dealers and suppliers rather than by the revered financial specialists (Wright, 1979). The perspective of the dominant coalition had legitimacy – all others were suppressed.

When dissenting viewpoints are discouraged an atmosphere of 'groupthink' emerges, narrowing the concerns and capacities of the organization (Janis, 1977). Major assumptions about customers and markets become institutionalized – that is, they are accepted as given (Zucker, 1977). And the indiscriminate embrace of such assumptions contributes to immoderation.

Slack Resources and Consensus

Long-successful organizations, because they are so rich in slack resources, are especially likely to produce consensus among their dominant coalitions and those employees and partners who share in the rewards (Meyer and Zucker, 1989, p. 23). There is enough wealth to satisfy everyone. But in companies beset by performance problems, owners may want to change to become more efficient.^[1] This will threaten the status and rewards of some constituents, causing them to challenge both strategies and underlying values. Such questioning can be a force for moderation.

Internal Darwinism

The gradual convergence of corporate culture on a few strategic ideals may be fostered by a kind of internal Darwinism (Burgelman, 1990; Miller, 1990, 1993a).

Take again the example of an innovative company. A firm's goals of technological leadership and its success with early inventions will encourage still more innovation. So additional scientists are recruited, promoted, and rewarded. The growing R&D group then embraces a 'think tank' culture that attributes success to innovation and presses for further novelty. Technological leadership soon becomes an even more dominant goal and leaders give R&D departments still more power and prestige. Before long, other departments and skills become less valued and less influential and the firm becomes increasingly focused on what it does well (Miller, 1990, 1993a; Miller and Chen, 1993a).

Such causal chains drive strategy-making processes to extremes – in this case of innovation, risk-taking and missionary R&D objectives. They promote a conformity that discourages any attempt to re-evaluate the environment or to gather the intelligence that might spur reorientation (Ashton, 1976; Campbell, 1970; Miller, 1990, 1993a; Miller and Chen, 1994).

Managers of troubled firms are more likely to have both the humility and the incentive to re-examine their biases and avoid extreme behaviour. A few major R&D flops, for example, may force them to re-evaluate the research department and its assumptions and to look elsewhere for inspiration (Hambrick and D'Aveni, 1988).

H2: Strategy-making processes will reach greater extremes after success than after mediocre performance. Success can cause companies to pursue tremendous innovation or total stability, to undertake rash risk-taking or adhere to hidebound conservatism, or to engage in extreme proactiveness or pure imitation.

INATTENTION: THE ATROPHY OF INTELLIGENCE

Success can discourage 'information processing' in organizations (Huber, 1991; Lant and Montgomery, 1987; Levinthal and March, 1981). Information processing may be characterized by the intensity and openness of communication among managers and workers, the sensitivity of scanning and control systems, the exchange and analysis of information, and the responsiveness of decisions and decision-makers to their environments ('adaptiveness') and to internal constraints ('integration') (Fredrickson, 1986; Miller and Friesen, 1984).

Many of the reasons why success can reduce information processing have already been given. For example, the cognitive factors we have discussed sharply circumscribe what the managers of thriving firms will scan for, pay attention to, and analyse (Lant, 1992; Levinthal and March, 1981; March, 1991). Political forces wed leaders of successful firms to particular strategies, make them hard to unseat, and leave them with little incentive to listen to the opinions of others (Pfeffer and Salancik, 1978). And, of course, intolerance and groupthink institutionalize the opinions of the dominant coalition while sapping legitimacy from the views of the disenfranchised (Miller, 1990, 1993a). This discourages attempts to question existing orientations and constrains the flow of ideas. There remain only a few additional factors that suggest a link between past success and a reduction in organizational intelligence.

The Extinction of Learning

Unless challenged repeatedly by problems, people and organizations fail to learn (Lant and Montgomery, 1987). Success ensures that the same kinds of stimuli will be targeted for attention and the same mental frameworks will be used to understand them (March, 1991; Starbuck and Hedberg, 1977). It also reduces the incentive to search for better ways of doing things (Levinthal and March, 1981). As a result, the managers of formerly thriving enterprises frequently fail to pick up signals that would suggest the need for change (Levitt and March, 1988; Weick, 1979). Indeed, in the absence of salient disappointments, administrators are free to engage in 'superstitious learning' in which they attribute good performance to whatever fits their predispositions (Levitt and March, 1988; March and Olsen, 1976). These convictions reduce the incentives for data collection and dissemination.

Focused Information Systems

Management information systems do not track the things managers believe to be unimportant or unchanging, but instead focus attention on what is thought to have mattered in the past (Starbuck, 1985; Staw et al., 1981). And in many successful businesses, executives develop the self-assurance to home in very precisely on what they believe explains their success. Their information systems then fix upon this and ignore everything else (Meyer and Starbuck, 1991). Such focused systems institutionalize and routinize gaps in organizational intelligence (Starbuck and Milliken, 1988; Wilensky, 1971). Problems, on the other hand, motivate managers to stray from their formal systems and to gather non-routine information (Kiesler and Sproull, 1982).

Success can also engender overconfidence among leaders who come to believe that they truly understand and know how to control their environments (Meyer and Starbuck, 1991; Starbuck and Milliken, 1988). Such notions make administrators reluctant to consult or listen to subordinates or to look at what their competitors and customers are doing (Kets de Vries and Miller, 1984).

H3: There will be a greater decline in information processing activity after success than after mediocre performance. Success causes companies to restrict the flow of communication, to allow scanning and control systems to deteriorate, to become more sloppy and less consultative in making decisions, to be less responsive to external forces and internal constraints, and to react only to crisis.

INSULARITY: SLUGGISH ADAPTATION TO ENVIRONMENT

The discussion so far suggests that organizations will be less able to adapt to their changing environments after success than after mediocre performance. Specifically, good performance creates inertia;^[2] it also leads to a general decline in information processing that insulates the firm from its environment and prevents top managers from noticing key external challenges.^[3]

The adaptation of an organization to the changes in its environment will be operationalized according to the normative propositions of the structural and strategic contingency literature.

Structural Contingencies

Contingency theorists for the past three decades have argued that managers must shape organizational structures and processes to fit the evolving context. Such adaptation is believed to be necessary for survival. Moreover, the contingency theorists are quite specific about the exact nature of the match required between organization and environment.

Reviews by Khandwalla (1977), Miller (1988) and Mintzberg (1979) show that the structural contingency literature identifies three dimensions of environment as having the greatest bearing on structure: they are uncertainty, hostility and heterogeneity.

Uncertainty refers to the level of change and unpredictability in client needs, competitive forces, and resources. Blau and Schoenherr (1971) and Burns and Stalker (1961) claim that increasing environmental uncertainty must be accompanied by more delegation of responsibility for routine decisions, more decentralization of authority for strategic decisions, and the empowerment of technocrats. These changes all increase the flexibility and expertise available to deal with growing complexity. Lawrence and Lorsch (1967) and Galbraith (1973) claim that uncertainty also requires increased organizational differentiation – that is, the incorporation of different time, task and interpersonal orientations in departments that need to deal with a large variety of contingencies. Finally, Galbraith (1973), Perrow (1971) and Thompson (1967) suggest that uncertainty necessitates more intensive lateral and vertical communications to effect co-ordination.

Hostility – the degree of competition and resource scarcity – also has implications for structure. Khandwalla (1973), Simons (1987) and Smith et al. (1991) note that hostility requires more elaborate and sensitive internal controls and more intensive internal communications to detect and formulate responses to competitive threats. It also demands greater depth and breadth of expertise in the form of expert managers or ‘technocrats’ (Mintzberg, 1979).

The heterogeneity of customer preferences and competitive challenges also has implications for structure. It is normally associated with decentralization of authority, and the differentiation of subunits (Galbraith, 1973; Rumelt, 1974; Thompson, 1967). This is said to allow greater sensitivity to different markets.

Organizations will be better at adapting to the changes in their environments after mediocre performance than after good performance. Specifically,

H4-1: The following relationships will be stronger after mediocre performance than after success:

- (a) Increasing uncertainty will induce more intensive internal communications, decentralization of strategy-making authority, more delegation of responsibility for routine decisions, reliance on trained technocrats, and more differentiation among departments.
- (b) Increasing hostility will induce more elaborate internal controls, more intensive internal communications, and greater reliance on technocrats.
- (c) Increasing heterogeneity will induce decentralization of authority and greater differentiation among sub-units.

Strategic Contingencies

The contingency approach has been extended to the field of business strategy. Managers, it is claimed, do not merely shape structure to fit the environment, they also adapt a firm's strategy-making processes (Fredrickson, 1986; Miles and Snow, 1978).

Compared to stable environments, uncertain ones in which tastes and products change quickly require more innovative product-market strategies as well as the risk-taking and proactiveness such strategies entail (Miller, 1988). Decisive updating can keep products lines relevant in a turbulent setting. Uncertainty is also said to require that administrators devote more effort to scanning their environments (Aguilar, 1967) and obtaining broad expertise ('multiplexity') in decision-making (Miller, 1987). Finally, uncertainty demands flexibility – that is, less reliance on precedent and tradition (Fredrickson, 1986).

Hostility also places demands on strategy-making process. Smith et al. (1991) believe that competition requires executives to do more to scan and gather information from their surroundings so as to be able to respond to threats more quickly and accurately. Khandwalla (1973) argues that competitive threats demand a broader, more 'multiplex' array of perspectives to better diagnose strategic attacks. Hostility is also said to require quicker responsiveness or adaptiveness to avoid punishment from rivals (Fredrickson, 1986; Miller, 1987).

As we expect success to decrease a firm's adaptiveness:

H4-2: The following relationships will be stronger after mediocre performance than after success.

- (a) Increasing uncertainty will induce more innovation, less conservatism, more proactiveness, more scanning, more multiplex decision-making, and a de-emphasis on traditions.
- (b) Increasing hostility will induce more scanning, more multiplex and adaptive decision-making, and less reliance on traditions.

METHOD

Our hypotheses compare the changes and adaptations that take place following success with those following mediocre performance. They concern significant transitions in structure and strategy-making process, and the extent to which these reflect parallel changes in the environment. This focus required that we examine periods of organizational history which could disclose such transitions.

The Sample

The only economical way to obtain rich details on the long-term evolution of many organizations and their environments is to use historical records in the public domain. Several years were spent by the researchers and their assistants in compiling the histories of 36 companies, each tracked for at least two decades. Firms were selected based on the availability of public information on the emergence of their strategies, processes, structures, and environments. Ours, then, is

not a random sample but one biased towards well-known businesses that survived for 20 years or more. In fact, the current data base updates and builds upon one we collected earlier (Miller and Friesen, 1980b).

Data sources included dozens of books and annual reports and hundreds of newspaper and magazine articles. These were used to map the histories of each firm, highlighting all the major decisions and events. The sample and some of the published data sources are listed in appendix 3. Firms were mostly medium and large sized, and were from a wide variety of industries. All had experienced periods of both success and mediocrity.

The Variables

The hypotheses determined our selection of variables, all of which are defined in appendix 4. Environmental variables included uncertainty, hostility, and heterogeneity, as per hypotheses 4-1 and 4-2.

Consistent with these hypotheses, structural variables included the use of formal controls, openness of internal communication, centralization of strategy-making authority, delegation of operating responsibility, use of expert technocrats, and differentiation (controls and communication may be considered as structural 'information processing' variables).

Strategy-making variables included aspects of 'information processing' highlighted by Fredrickson (1986) and Miller (1987). These included environmental scanning and decision analysis, the multiplexity (breadth) of perspectives brought to bear in decision-making, the fit of decisions with internal constraints (integration), and managerial time horizons (futuraity). Other process dimensions taken from Mintzberg (1973) and Miller (1987) included reliance upon precedent or tradition in making decisions, product market innovation, risk-taking, proactiveness, and the codification or explicitness of strategy.

All the above variables were rated at the beginning and end of each period of analysis (to be defined below) along a 7-point scale. A score of 1 indicated that a firm rated 'much lower' along the variable than other firms in the sample, 4 represented about average, and 7 indicated 'much higher or more' than other firms. Performance was also measured (see below), as were aspects of context such as the duration of the period of analysis, its date, the age of the firm, and whether there was a change in CEO.

Partitioning and Scoring the Historical Data

It was necessary to partition the lengthy histories that we compiled into intervals across which to measure the changes and adaptations of our hypotheses. Although there is no one best way to define such periods, our earlier work provided some guidance. Miller and Friesen (1984) have shown that organizational transitions come about when a destabilizing decision or event creates imbalances or incongruities among environment, structure and process. They found that the initial change, usually a significant shift in strategy, context or structure, is often followed by a cascade of compensatory adjustments aimed at restoring equilibrium. The organization then returns once more to a state of relative calm.

There are many instances of such transitions. Product market diversification, for example, may create administrative burdens that require restructuring in the

form of better controls, profit centres, decentralization, and more planning. To take another case, increasing uncertainty in the environment may demand more innovation and flexibility; this then requires more intensive internal communication, more delegation of authority to experts, and more collaborative decision-making (Miller and Friesen, 1980b, p. 271).

Such examples reveal how an initial change can give rise to a cascade of subsequent adjustments. Our hypotheses concerning organizational evolution and adaptation can only be tested by capturing such fundamental transitions. We set out, therefore, to assess the changes that occur between the time just before the destabilizing event and the restoration of tranquility.

Although most such transitions take years to occur, some earthshaking revolutions happen in a matter of months (see, for example, Chandler, 1962; Hinings and Greenwood, 1988; Miller and Friesen, 1984; Pettigrew, 1973; Rumelt, 1974). We could not therefore standardize the interval over which to measure change. Too short a period would capture nothing of relevance to our hypotheses. Too long a period would conceal opposite changes that cancel each other. Thus the length of the periods was allowed to vary. Raters were told to begin or end a period during a tranquil interval just before (1) a change in *leadership* (i.e. a new CEO); (2) creating a new *strategy* (e.g. introducing a significant new product, entering a major new market, adopting a new production or process technology, using a very different marketing approach, or making a significant merger or acquisition); or (3) designing a new *structure* (e.g. materially modifying the power distribution or reporting relationships, imposing a new set of control, planning and reporting procedures, establishing an important new department). A new period could also be triggered by (4) a major change in the external *environment* (e.g. a powerful new competitor, a new dominant design or technology, deregulation). These four categories of leadership, strategy, structure and environment were selected as they represent perhaps the most frequently studied causes and effects of organizational change (Miller and Friesen, 1984; Tushman and Romanelli, 1985).

Significant adaptation and change in structure and process can often take four or more years (Chandler, 1962; Grinyer et al., 1988; Hinings and Greenwood, 1988; Pettigrew, 1973; Rumelt, 1974). Therefore, once a period was begun, it was continued for at least that long to allow something of consequence to happen. Sometimes, however, firms were in a holding pattern, undergoing essentially no changes in strategy, structure or environment within that interval. Then periods were extended until any four variables (including those of environment or context) altered. This was done to avoid having too many uneventful periods.

In order to prevent opposite changes from cancelling each other, periods were terminated even before four years had elapsed if any change reversed itself (e.g. if power, which was becoming more widely distributed, suddenly is recentralized). The failure to do this would have resulted in an unacceptably great loss of information. The end of one period always coincided with the beginning of the next.

Variables of environment, structure and strategy-making were scored on the 1–7 scales at the beginning and end of the periods. There were 165 such points in time which gave rise to 165 profiles of scores along the 20 variables. These

yielded 129 periods of transition (165 less the 36 initial profiles for each firm) which were characterized by the *differences* between the scores at the beginning and end of each period. Periods had a mean duration of 6.3 years, with a standard deviation of 3.7 years.

For a randomly selected half of the histories, both period boundary definition and variable scoring were performed by two independent raters working double-blind. Each had doctoral training in management policy and organizational theory and had read at least 20 histories of a diverse assortment of companies in order to have a broad enough basis of comparison for assigning ratings. Raters were instructed to score the periods only for a firm's dominant business.

In 86 per cent of the cases, the raters agreed upon the period boundaries; when there was disagreement, the shorter period was used to avoid obscuring offsetting changes. In virtually all cases, the same firms exhibited both star (post success) and non-star (post mediocre) periods.

Raters were given detailed definitions and examples to use in scoring the variables. Interrater reliabilities averaged 0.87 for the 65 profiles rated. Only in 4 per cent of the cases did scores differ between raters by more than 2 points on the 7-point scales. Scores were averaged in cases of disagreement. Missing data accounted for 4 per cent of the observations.

Assessing Performance

Because performance is such a critical aspect of this research, it was measured in two stages. First, success was measured by estimating average annual growth in sales and after-tax profitability (ROI, or, where this was unavailable, return on sales) for each period of analysis. This information was obtained from published sources such as financial reports, Value Line, 10K statements, and tabulations in books and articles.

Armed with the performance data, we then consulted industry sources and descriptions in the published data base to assess how a firm compared to its contemporary competitors. A period could only be classified as 'post-success' if both average growth rate and profitability were higher than the industry average for at least the last three years of the *preceding* period. Current performance was irrelevant for this classification. Failing the availability of that average, the performance of leading competitors was used as the basis for comparison. Because our hypotheses concern the impact of long-term unambiguous success, if a firm's performance was clearly superior to its competitors' along growth but not profitability, or *vice versa*, the period was dropped from the study.

There are two performance variables related to each period. The first is performance in the prior period – the critical variable that classifies a period as post-success or post-mediocre and upon which our hypotheses rest. The second is performance during the current period. Note that in the subsequent analyses, performance is a dummy variable. Firms are simply classified as stars (1) or non-stars (0). Given the variety of industries and time periods, this was the most conservative way to proceed. Small gradations of difference in performance would be unreliable in so diverse a context of industries and periods.

Although we started with 129 periods of analysis, we lost 17 because of the ambiguity or inadequacy of performance data, leaving a sample of 112 periods; 48 periods that followed success, and 64 that followed mediocrity.

Validity Test Questionnaires

To guard against common method variance and ensure the validity of the data, we polled the top executives (presidents, CEOs or chief operating officers) of 12 of the 36 firms – those who had not yet retired and were with the firm at the vice presidential level or higher during at least one period of analysis. Ten of these executives responded. Executives were asked to complete a questionnaire gauging the changes that took place along all of our variables during the most recent period for which we had data. The Spearman coefficient was deemed too undemanding to assess inter-rater reliability as it would allow very high r 's even when raters disagree on the direction of change – a very serious disagreement given the nature of our hypotheses. So we looked in more detail at rater correspondence.

On the 1 to 7 scale, where 1 represented a great decrease in a variable, and 7 a great increase, respondents agreed 85 per cent of the time in the direction of change. Slight differences in magnitude such as 6 *vs.* 7 or 2 *vs.* 3 were not considered important given that executives did not have the same broad basis for comparison as the raters. But there were two types of significant disagreements: one respondent might report no change (4) and another a notable increase or decrease; or respondents might disagree on the direction of change. The former disagreement occurred for 13 per cent of the ratings, the latter for only 2 per cent. The change data thus had an acceptable level of validity.

The Company Effect

The unconventional nature of our data made it necessary to exercise great care in testing our hypotheses. Because there are an unequal number of periods per firm, we were concerned that a 'firm' variable could bias our results. To rule this out a categorical variable was defined with 36 categories – one per firm. Then we ran ANOVAs on all of the performance and dependent change variables to determine if 'firm' had any effect on the pattern of evolution. Only one of the 20 analyses reached statistical significance – that for change in the use of technocrats. No other results were significant even at the 0.30 level. These analyses are available on request.

The Serial Correlation Effect

Because there are multiple periods per company, it was necessary to ensure that our analyses are not contaminated by autocorrelation or heteroscedasticity. Plots of regression residuals by firm revealed no such problems. To test for bias in assessing hypothesis 1, predicted values were used as instruments for the lagged dependent variables (the stochastic regressors). Results were consistent with table I.

The subsequent analyses use transitions or *change scores* in the variables, not raw scores, where change is calculated as the difference between the scores at the beginning and the end of a period. The serial correlations for these change variables were very modest indeed. In fact, the average interperiod lagged correlation for the changes was only -0.13 , a *negative*, non-significant value accounting for less than 2 per cent of the variance. None of the 17 organizational change variables correlated positively and significantly over time at even the 0.10 level. The Durbin statistics were all non-significant. Clearly, the

transition observations used to test hypotheses 3, 4-1 and 4-2 are quite independent.

The Period Duration Effect

Because periods of analysis are not of equal duration we took pains to establish how this variation might influence the findings. First, we compared the durations of the periods for the post success and post mediocrity subsamples. Durations were almost identical (6.31 *vs.* 6.34 years, respectively), and did not differ at even the 0.80 level of significance.

Also, every one of the analyses used to test the hypotheses was run four times. The first analyses incorporated the total sample of 112 periods; these results are reported in tables I to IV-2. To establish the robustness of these findings we dropped all periods of longer than 6 years from the sample, leaving 75 periods with a mean duration of 4.2 years, and a standard deviation of 1.3. As can be seen from the results summary of appendix 5, the findings for these analyses often supported the hypotheses even more than those for the total sample, in spite of the fewer degrees of freedom. Hence, it was most conservative to present the results for the total sample rather than just those for the shorter periods.

The last few years of long periods can be so far removed from prior bouts of success that they may hardly differ from intervals following mediocre performance. Short periods, on the other hand, are on average closer to the interval of success, and will be more influenced by the forces described in our hypotheses. Hence more support for our hypotheses was expected in shorter than in longer periods.

To verify this we performed a median split of the sample according to period duration and reran all analyses on the two subsamples. The first subsample included only periods of 5 years or less (*N* of 63, mean of 3.8 yrs, s.d. of 1.1), the second subsample included all other periods (*N* = 49, mean = 9.5 yrs, s.d. 3.4). Appendix 5 confirms that the hypotheses were much more strongly supported in the subsample of shorter periods. And again the findings of the shorter period very much replicate those of the total sample. All of these analyses are summarized in appendix 5 and are available upon request.

As a further control for the effect of period duration on our results, the duration was included as an independent variable in all of our multivariate analyses. Duration was shown to have an impact on the dependent variables studied in only 2 of 48 analyses, a finding consistent with a purely chance effect.

FINDINGS

Appendices 1 and 2 present the descriptive statistics and correlation matrix. It is interesting that the former reveals very few differences between past stars and mediocre firms. The stars were slightly more innovative, less adaptive, more apt to refer to tradition and precedent in making decisions, and they tended to codify their strategies more explicitly. These differences were very much in line with hypotheses 1 and 2. There were absolutely no differences, however, between the environments or contexts of the two subsamples.

Controls

Hypotheses 1 to 4 examine periods of organizational history and the changes that occur in environments and organizations over time. But there are numerous factors that we have not said much about that can influence organizational evolution and so must be controlled for when testing our hypotheses. As noted in hypotheses 4-1 and 4-2, environmental uncertainty and hostility can influence most of the dependent variables of structure and process. Inertia, for example, may be less likely in a changing or competitive environment than in a stable, unchallenging one (Meyer and Starbuck, 1991). It was also vital to control for the length of the period of analysis, since the longer the interval, the more change is likely to take place. Date can also be a factor, with structures and processes having been more primitive and slower to evolve earlier in the century than at present (Chandler, 1962). Furthermore, the age of an organization is said to be positively associated with inertia (Miller and Chen, 1994). In contrast, a new CEO may be an impetus to organizational responsiveness and renewal (Tushman et al., 1989). Finally, current success may be correlated with past success (the crucial predictor), confounding the relationships of our hypotheses. All of these factors: uncertainty, hostility, period duration, date, age, CEO succession (a 0-1 dummy variable), and current performance (another 0-1 dummy), are controlled for in examining the relationships between prior success and organizational evolution.

The numerous control variables called for the use of multiple regression analyses. Each of the hypotheses was even more strongly supported, however, in pared down regressions that did not incorporate the controls (these analyses are all available from the author).

Hypothesis 1: Success and Adherence to the Status Quo

According to hypothesis 1, periods following good performance will exhibit more inertia than those following mediocre performance. Such inertia can be represented by the 'lagged' or serial correlations of a variable with itself. Given the scope of the hypothesis, all organizational variables were included in these analyses.

To test hypothesis 1 we regressed each structure and process variable at the end of the period on the corresponding variable at the beginning of the period. The products of the dummy for past performance and past structure or process were used as interaction or 'moderator' variables to establish whether the lagged relationships were higher in the post-success subgroup than in the mediocre subgroup. A significant and positive interaction would indicate that the correlation between past and current orientation was highest in the post-success subgroup. Separate regressions were run for each of the 17 organizational variables, and each analysis controlled for environmental uncertainty, hostility, the dummy for current performance, CEO succession, period duration, date, and age.

Table I indicates that a majority of these interaction terms – 10 of the 17 – were positive and significant, 7 at beyond the 0.01 level, 1 at beyond the 0.05 level, and 2 at beyond the 0.10 level. Only two terms were negative, both non-significant. Hence hypothesis 1 receives modest support.

Because endogenous stochastic (lagged) variables can make OLS regression analysis unreliable, we tried to verify the findings using simple correlations. The

Table I. Current structure and process regressed on past structure and process

Standardized betas	Controls	Current structure on past structure				
		Commun'n	Cent'n	Deleg'n	Technoc.	Diff'n
Past struct.	0.42**	0.24	0.14	0.27**	0.82***	0.73***
Past performance dummy	-0.28*	-0.90*	-0.45	0.87***	-0.02	0.05
Past struct. × past perf.	0.08	0.65+	0.35	0.85***	-0.04	-0.07
Uncertainty	0.06	0.00	0.03	0.12	0.12	0.13
Hostility	0.26*	0.25+	0.16	-0.07	-0.01	0.02
Current perf.	0.31***	0.32**	0.00	0.14	-0.01	0.03
Succession	-0.12	-0.02	-0.45**	0.23**	0.01	0.02
Duration	0.03	0.17	0.09	0.10	0.10	-0.08
Date	0.06	0.08	-0.05	-0.07	0.14**	0.09
Age	0.08	-0.08	-0.06	0.00	-0.06+	0.03
R-squared	0.45	0.33	0.29	0.39	0.88	0.70
Overall F	6.53	3.39	3.97	5.57	70.13	23.59
p-value	0.000	0.001	0.000	0.000	0.000	0.000
(d.f. 11, 100)						
Standardized betas	Proact.	Innov.	Current process on past process			Multip.
			Risk	Scann.	Analy.	
Past process	0.09	0.08	0.21*	0.30	0.32	0.06
Past performance dummy	-0.78**	-0.10	-0.78**	-0.86**	-0.52*	-0.94***
Past process × past perf.	0.88**	0.21	0.96***	0.67*	0.34+	0.89***
Uncertainty	0.30**	0.40**	0.15	0.12	0.01	0.04
Hostility	0.05	0.04	0.08	0.16	0.33**	0.25*
Current perf.	0.17	0.24*	0.03	0.32**	0.28**	0.34**
Succession	-0.12	-0.12	-0.09	-0.02	0.11	-0.13
Duration	0.12	0.12	0.06	0.02	-0.02	-0.02
Date	-0.06	0.09	0.07	0.00	-0.04	0.00
Age	-0.03	-0.10	-0.02	0.09	0.04	0.09
R-squared	0.32	0.34	0.34	0.47	0.31	0.35
Overall F	4.78	5.17	5.05	7.19	4.01	5.39
p-value	0.000	0.000	0.000	0.000	0.000	0.000
	Integ'n	Futur.	Current process on past process			Trad.
			Explic.	Adapt.		
Past process	0.16	0.04	0.16+	0.05	0.21+	
Past performance dummy	-0.97***	-0.77**	-1.05**	-0.35	0.04	
Past process × past perf.	0.87**	0.84**	1.19***	0.14	0.14	
Uncertainty	0.03	0.23*	-0.03	0.10	-0.30**	
Hostility	0.15	0.12	0.02	0.26	-0.01	
Current perf.	0.34**	0.06	0.19+	0.47**	-0.05	
Succession	-0.09	-0.14	-0.25**	-0.15	-0.12	
Duration	-0.02	-0.04	-0.05	0.11	0.08	
Date	-0.11	0.04	0.01	-0.04	-0.05	
Age	0.07	0.02	0.07	0.05	0.05	
R-squared	0.31	0.25	0.30	0.27	0.23	
Overall F	4.32	3.21	4.18	3.74	2.97	
p-value	0.000	0.001	0.000	0.000	0.003	

+, *, **, *** indicate significance levels of 0.10, 0.05, 0.01 and 0.001

results were even stronger when we compared only the simple lagged correlations of post-success and mediocre subgroups, with the former having an average lagged correlation of 0.62 and the latter one of 0.31.^[4] Post-success periods also had significantly higher Fisher Z -transformed r 's than their mediocre counterparts at the 0.05 level in 13 of 17 comparisons.

It is noteworthy that hypothesis 1 was more convincingly supported for strategy-making process (with 8 of the 11 interactions significant) than for structure (with only 2 of the 6 interactions significant). Structural variables such as differentiation, the distribution of power, and the use of experts in decision-making may be much harder to change than process variables, even for mediocre past performers. Such structural variables are tied both to inertial political considerations and longstanding industry requirements (e.g. there is greater differentiation and more technocrats in computer firms than in steel mills). Process variables are easier to change, however, and here, post-success firms stay put while their mediocre counterparts evolve.

Hypothesis 2: The Development of Extremes

According to hypothesis 2, periods following good performance are more likely to show firms reaching certain extremes than are periods following mediocre performance. During the former, firms will be buoyed by confidence and consensus to fully pursue the product-market orientations they believe in. Whereas hypothesis 1 pertained to all aspects of structure and process, the second hypothesis refers only to product-market innovation, risk-taking and proactiveness.^[5]

Extremes were defined as scores of 1 (very low relative to other firms) or 7 (very high) on the 7 point-scale. A dummy variable was defined that would take the value of 1 when extreme, and 0 otherwise. This dummy was regressed against past performance and all of the control variables.

Table II reveals that post-success periods exhibited more extremes than their mediocre counterparts for all three variables ($p < 0.01$ for 2 analyses, $p < 0.05$ for the third). To establish the robustness of this finding, we reran the analyses defining extremes as scores of 2 and 6 as well as 1 and 7. The findings were replicated. Appendix 1 shows that these results were in no way attributable to differences in the subsample means. It also reveals that the standard deviations of the variables are higher among post-success periods than others, again supporting the notion of greater extremes.

In short, post-success firms are particularly likely to reject strategic nuances in favour of the black and white. They are more apt to take many risks or to eschew them entirely, to act without regard to competitors' practices or resort to pure mimicry of them, and to engage in innovative excess or stick resolutely to an established line.

Hypothesis 3: Declining Information Processing

According to hypothesis 3 information processing activity will decline more after good performance than after mediocre performance. Information processing variables encompass those of both structure and process. The former include control and communication systems; the latter scanning, analysis, multiplexity, integration, futurity, and adaptiveness. Again multiple regression analyses were run, this time taking as the dependent variables *change* during the period in information

Table II. Extremes regressed on past performance and controls

<i>Standardized betas</i>	<i>Proact.</i>	<i>Risk</i>	<i>Innov.</i>
Past performance dummy	0.25**	0.23**	0.17*
Uncertainty	0.19+	0.15	0.28*
Hostility	-0.08	-0.12	-0.17
Current perf.	-0.06	-0.28**	-0.02
Succession	-0.06	0.09	-0.05
Duration	0.03	-0.13	-0.03
Date	0.09	-0.07	0.03
Age	-0.15	-0.06	-0.18
R-squared	0.13	0.17	0.14
Overall F	1.92	2.59	2.11
p-value	0.065	0.013	0.041
(d.f. 9, 102)			

+, *, **, *** indicate significance levels of 0.10, 0.05, 0.01 and 0.001

processing. Change was measured as the difference between the variable scores at the beginning and the end of the periods of analysis. The past performance dummy was the primary predictor variable and the usual controls were entered into the regressions.

According to table III, post-success periods exhibited greater declines in intelligence than those following mediocre performance. For 7 of the 8 information processing variables, these differences were statistically significant at beyond the 0.001 level. For the futurity (time horizon) variable, the significance level was only 0.10.

It is interesting that *current* success had an opposite, positive relationship to changes in information processing. Clearly, it is past, not current performance that is associated with a decline in organizational intelligence activity. Indeed, it may be that although past success can erode vigilance, vigilance might contribute positively to current performance.

Hypothesis 4: Adapting the Organization to its Environment

Hypothesis 4 and its components predicted that the normative propositions of the structural and strategic contingency theorists would be more strongly supported after mediocre performance than after success. This was tested in tables IV-1 and IV-2 by regressing change in structure and process against change in environment, where change was measured as the difference in variable scores between the beginning and end of the periods. These analyses incorporate the usual control variables, the dummy for past performance, and a variable that is the product of that dummy and the change in environment. If the last interaction term is significantly negative, then the relationship between change in environment and change in structure or process is weaker after success than after mediocre performance, as predicted. The sections of tables IV-1 and IV-2 correspond to each of the parts of hypothesis 4-1 and 4-2 respectively.

Structural relationships: On table IV-1 the hypothesized changes in structure are regressed on environmental, past performance, interaction and control variables.

Table III. Change in information processing regressed on past performance and controls

<i>Standardized betas</i>	<i>Scanning</i>	<i>Controls</i>	<i>Commun'n</i>	<i>Analysis</i>	<i>Multiplex</i>
Past performance dummy	-0.39***	-0.41**	-0.54***	-0.31***	-0.41***
Uncertainty	0.15	0.06	0.10	0.01	-0.03
Hostility	0.03	0.29**	0.23+	0.31**	0.22+
Current perf.	0.24*	0.27*	0.22+	0.25*	0.20*
Succession	0.20+	-0.02	0.06	0.24*	0.01
Duration	-0.05	-0.07	0.05	-0.11	-0.11
Date	-0.02	-0.15	0.05	-0.16	-0.11
Age	-0.04	-0.05	-0.03	-0.04	-0.03
R-squared	0.21	0.23	0.35	0.22	0.20
Overall F	2.79	3.00	4.78	3.17	3.27
p-value	0.009	0.005	0.000	0.003	0.002
(d.f. 9, 102)					

<i>Standardized betas</i>	<i>Adapt.</i>	<i>Integr'n</i>	<i>Futur.</i>
Past performance dummy	-0.45***	-0.39**	-0.13+
Uncertainty	0.22*	0.07	-0.09
Hostility	0.25*	0.11	0.21*
Current perf.	-0.21*	-0.17	-0.07
Succession	-0.02	0.02	-0.03
Duration	-0.04	-0.14	-0.01
Date	-0.16	-0.20+	0.04
Age	-0.08	-0.07	0.13
R-squared	0.27	0.20	0.08
Overall F	4.81	3.19	1.06
p-value	0.000	0.003	0.40
(d.f. 9, 102)			

+, *, **, *** indicate significance levels of 0.10, 0.05, 0.01 and 0.001

Seven of the 10 interactions are negative and significant at beyond the 0.05 level. Although there was only sporadic support for the relationships of structure with uncertainty, the findings for hostility and heterogeneity were strong and unanimous.

Strategy-making process relationships: On table IV-2 the hypothesized changes in process are regressed on the same variables as in table IV-1. Again, 7 of the 10 interactions are negative and significant, 6 at beyond the 0.05 level, one at beyond the 0.10 level.

It seems that after success, organizations are indeed less able or less inclined to adapt their structures and processes to changes in environment than after mediocre performance. And the earlier findings on inertia, extremes, and information processing help to explain why.

Performance Implications

Given the past success was associated with considerable inertia, strategic extremes, declining intelligence, and a failure to keep up with the evolving environment, we suspected that it might also lead to deteriorating performance.

Table IV-1. Changes in structure regressed on changes in environment and controls

Standardized betas	(a) Changes in structure on changes in uncertainty				
	Commun'n	Cent'n	Deleg'n	Technoc.	Diff'n
Chg. uncertainty	0.02	0.10	0.24*	0.26*	0.24
Past performance dummy	-0.51***	-0.06	-0.21*	-0.19*	0.08
Chg. unc. × past perf.	-0.14	-0.26*	-0.09	-0.08	-0.23*
Uncertainty	0.11	-0.06	0.00	0.01	0.04
Hostility	0.26*	0.14	-0.03	0.12	0.08
Current perf.	0.25*	-0.10	0.25*	0.07	0.14
Succession	0.08	-0.33**	0.19+	0.03	0.06
Duration	0.07	0.07	0.03	0.21*	-0.06
Date	0.05	0.05	0.01	0.04	0.01
Age	-0.03	0.01	-0.05	-0.10	-0.01
R-squared	0.36	0.19	0.15	0.15	0.06
Overall F	3.92	2.28	1.59	1.73	0.68
p-value	0.000	0.019	0.124	0.085	0.74
(d.f. 11, 100)					

	(b) Changes in structure on changes in hostility		
	Controls	Commun'n	Technoc.
Chg. hostility	0.45*	0.35*	0.16
Past performance dummy	-0.44***	-0.54**	-0.11*
Chg. host × past perf.	-0.25*	-0.32*	-0.30*
Uncertainty	0.05	0.14	0.08
Hostility	0.11	0.11	0.14
Current perf.	0.34*	0.31*	0.08
Succession	-0.08	0.05	0.05
Duration	-0.07	0.00	0.22*
Date	-0.11	0.06	0.00
Age	0.01	0.03	-0.09
R-squared	0.29	0.42	0.16
Overall F	3.15	4.94	1.79
p-value	0.002	0.000	0.072

	(c) Changes in structure on changes in heterogeneity	
	Central'n	Differ'n
Chg. heterogeneity	0.10	0.83**
Past performance dummy	-0.02	-0.14
Chg. hetero. × past perf.	-0.33*	-0.46***
Uncertainty	-0.03	0.06
Hostility	0.06	0.05
Current perf.	-0.13	0.11
Succession	-0.33**	0.05
Duration	0.05	-0.06
Date	0.05	-0.07
Age	-0.07	0.02
R-squared	0.20	0.39
Overall F	2.48	6.28
p-value	0.011	0.000

+, *, **, *** indicate significance levels of 0.10, 0.05, 0.01 and 0.001

Table IV-2. Changes in process regressed on changes in environment and controls

	<i>Proact.</i>	<i>(a) Changes in process on changes in uncertainty</i>				<i>Trads.</i>
		<i>Innov.</i>	<i>Risk</i>	<i>Scan</i>	<i>Multip.</i>	
Chg. uncertainty	0.24*	0.34**	0.37**	0.05	0.13	-0.38**
Past performance dummy	-0.14	-0.14	0.08	-0.36***	-0.39***	0.11
Chg unc. × past perf.	-0.32**	-0.26*	-0.30*	-0.14	-0.10	0.22*
Uncertainty	-0.01	-0.06	-0.09	0.15	-0.06	-0.09
Hostility	0.10	0.15	0.06	0.05	0.22+	-0.07
Current perf.	0.21+	0.20+	-0.03	0.25*	0.21*	-0.06
Succession	0.06	0.04	-0.01	0.22*	0.00	-0.23*
Duration	0.12	0.11	0.14	-0.04	-0.12	0.08
Date	0.01	0.04	0.11	-0.02	-0.11	0.02
Age	0.12	0.11	0.13	-0.04	-0.03	-0.14
R-squared	0.15	0.16	0.14	0.22	0.21	0.25
Overall F	1.77	1.89	1.56	2.31	2.68	3.28
p-value	0.076	0.055	0.129	0.019	0.006	0.001

(d.f. 11, 100)

	<i>Scan</i>	<i>(b) Changes in process on changes in hostility</i>		
		<i>Multiplex</i>	<i>Adapt.</i>	<i>Trads.</i>
Chg. hostility	0.39*	0.47**	0.29*	-0.30
Past performance dummy	-0.49***	-0.47**	-0.48***	0.15
Chg. host × past perf.	-0.13	-0.23*	-0.17+	0.21*
Uncertainty	0.17	-0.02	0.23*	-0.20
Hostility	-0.18	0.02	0.14	0.04
Current perf.	0.32**	0.27*	0.25*	-0.10
Succession	0.19*	-0.05	-0.02	-0.26**
Duration	-0.07	-0.12	0.04	0.07
Date	0.00	-0.08	-0.15	0.03
Age	0.02	0.04	-0.04	-0.16
R-squared	0.28	0.26	0.30	0.22
Overall F	3.01	3.53	4.17	2.79
p-value	0.003	0.001	0.000	0.004

+, *, **, *** indicate significance levels of 0.10, 0.05, 0.01 and 0.001

Unfortunately, this was not a hypothesis that could be tested by this research. The universal phenomenon of regression towards the mean implies that good performers are more likely to experience declines in performance than poor performers. Hence our exploration of this issue must be extremely tentative.

We found the classification of a period as 'post-success' was negatively correlated with being classified as successful in the following period ($r = -0.12$, $p = 11$). In other words, former stars had a slightly lower probability of being stars in the present period than even their mediocre counterparts. Prior success does nothing to help subsequent success; indeed, the findings, though tentative, suggest that the opposite is true. Appendix 1 shows that the performance gap between stars and average performers completely disappears within one period.

CONCLUSION

Our findings indicate that past performance has a great effect on the way in which businesses evolve. Compared to periods following mediocre or poor performance, those following success (a) exhibit more inertia in process, and, to a lesser degree, in structure; (b) they display more extremes of risk-taking, proactiveness and innovation; (c) they evidence a greater decline in information processing activity; and (d) they exhibit poorer fit with their environments. Perhaps because of this, their initial performance edge vanishes.

We would like to stress that the findings reported here are conservative in two ways. First, they are based on all of our periods of analysis, even though the findings from comparing post-success and post-mediocrity periods of 6 years or less show even stronger support for the hypotheses (see appendix 5). Second, we have bifurcated the sample according to performance. It is reasonable, however, given the rationales behind our hypotheses, that the contrasts between the subsamples would be even greater when comparing periods following extreme success with those following extreme failure.

Our findings are consistent with the notion that different modes of evolution apply in different circumstances. For example, past success may give rise to periods of 'convergence' (Tushman and Romanelli, 1985) or 'momentum' (Miller and Friesen, 1980a, 1984) as it encourages ideological and strategic continuity and a honing of the structures and processes that support it. It can often be a prelude to inertia, immoderation, inattention, and insularity. Poor performance, on the other hand, makes it more likely that self-evaluation will take place and that existing values and strategies will be challenged. This can pave the way for revival, restraint, reawakening, and realignment with the environment.

Perhaps there is a dialectical process at work in organizations that stems in part from the effects of past performance. Fine performance gives rise to periods of inertia or convergence, which can lead to excesses. These may sometimes result in difficulties that stimulate firms to become more adaptive.

This research must, of course, be viewed as suggestive rather than definitive. First, its sample is biased since it includes only companies significant enough to be written about extensively in the public domain. Second, it encompasses a heterogeneous mix of industries, nations and time periods. And third, it does not allow the refined assessment of many aspects of product market strategy or organization structure.

We urge subsequent investigators to continue to examine how organizations evolve over time. They might wish to extend this study by looking at the content of strategy as well as the process, and by confining their research to a single industry and time period that they can understand in greater depth. It might also be useful to continue to explore alternative ways of dividing organizational histories into standardized periods of analysis. This would necessitate a better understanding of the periodicity of various forms of organizational change. It is quite conceivable, for example, that organizational process changes more quickly than structure, and that the duration of periods of analysis would have to be tailored to the domain of the study.

APPENDIX 1: MEANS SPLIT ON THE BASIS OF PAST SUCCESS

	<i>Post-success</i> <i>N = 48</i>		<i>Post-mediocre</i> <i>N = 64</i>		
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>p-value</i>
STRUCTURE					
Controls	3.98	1.90	4.27	1.66	NS
Communications	4.22	1.57	4.46	1.40	NS
Centralization	5.38	1.62	5.66	1.12	NS
Delegation	5.15	1.44	5.11	1.38	NS
Technocratization	3.21	1.88	2.98	1.40	NS
Differentiation	3.92	1.25	4.20	1.59	NS
STRATEGY-MAKING PROCESS					
Proactiveness	4.81	1.67	4.44	1.36	NS
Innovation	4.81	1.45	4.30	1.20	0.05
Risk-Taking	4.54	1.64	4.09	1.40	NS
Scanning	4.38	1.44	4.67	1.36	NS
Analysis	3.93	1.64	4.23	1.56	NS
Multiplexity	4.00	1.52	4.23	1.41	NS
Integration	4.25	1.64	4.51	1.28	NS
Futurity	4.50	1.44	4.41	1.35	NS
Explicitness	5.52	1.38	5.11	1.19	0.06
Adaptiveness	4.58	1.53	4.91	1.26	0.10
Traditions	4.17	1.66	3.75	1.60	0.09
ENVIRONMENT					
Uncertainty	4.07	1.23	3.89	1.29	NS
Hostility	4.49	1.49	4.50	1.74	NS
Heterogeneity	3.54	1.41	3.72	1.83	NS
CONTEXT					
Current perf. dummy	0.42	0.50	0.53	0.50	NS
Succession dummy	0.38	0.49	0.34	0.48	NS
Pd. duration (yrs)	6.31	3.95	6.34	3.57	NS
Date (year)	1941	33	1943	34	NS
Age of firm (yrs)	34.3	20.4	35.6	18.3	NS

APPENDIX 2: PEARSON CORRELATION MATRIX

	1	2	3	4	5
1. Uncertainty	-1.000				
2. Hostility	0.450	1.000			
3. Heterogeneity	0.418	0.227	1.000		
4. Controls	0.241	0.189	0.167	1.000	
5. Communications	0.001	0.046	-0.130	0.711	1.000
6. Centralization	0.006	0.034	-0.051	0.190	0.206
7. Delegation	0.121	-0.018	0.003	0.082	0.257
8. Technocrats	0.553	0.112	0.501	0.308	0.135
9. Differentiation	0.455	0.245	0.732	0.313	-0.013
10. Proactiveness	0.311	0.027	0.134	0.215	0.193
11. Innovation	0.413	0.041	0.166	0.196	0.299
12. Risk-taking	0.308	0.066	0.264	-0.093	-0.022
13. Scanning	0.208	0.246	0.031	0.658	0.661
14. Analysis	0.191	0.244	0.069	0.686	0.616
15. Multiplexity	0.160	0.137	0.057	0.745	0.676
16. Integration	0.024	0.005	-0.250	0.581	0.622
17. Futurity	0.340	0.139	0.254	0.487	0.369
18. Explicitness	-0.101	-0.090	-0.137	0.288	0.376
19. Adaptiveness	0.071	0.028	0.010	0.636	0.699
20. Traditions	-0.343	-0.125	-0.230	-0.152	-0.154
21. Past success	0.068	-0.003	-0.053	-0.082	-0.081
22. Current success	-0.170	-0.453	-0.068	0.271	0.399
23. Succession	0.140	0.303	0.015	-0.067	0.014
24. Duration	-0.071	-0.091	-0.134	0.210	0.241
25. Date	0.477	0.311	0.449	0.356	0.075
26. Age	0.144	0.280	0.012	0.298	0.049

	6	7	8	9	10
6. Centralization	1.000				
7. Delegation	-0.286	1.000			
8. Technocrats	-0.020	0.304	1.000		
9. Different'n	-0.091	0.148	0.437	1.000	
10. Proactiveness	0.283	0.125	0.203	0.308	1.000
11. Innovation	0.179	0.157	0.342	0.252	0.728
12. Risk-taking	0.113	-0.083	0.197	0.202	0.648
13. Scanning	0.226	0.065	0.140	0.103	0.257
14. Analysis	-0.002	0.214	0.257	0.174	0.146
15. Multiplexity	0.087	0.165	0.218	0.153	0.165
16. Integration	0.213	0.020	0.104	-0.180	-0.005
17. Futurity	0.295	0.053	0.298	0.358	0.574
18. Explicitness	0.315	0.002	-0.053	-0.084	0.167
19. Adaptiveness	0.201	0.227	0.143	0.114	0.328
20. Traditions	-0.031	-0.105	-0.218	-0.305	-0.486
21. Past success	-0.103	0.013	0.069	-0.098	0.124
22. Current success	0.052	0.063	-0.008	-0.181	0.183
23. Succession	-0.393	0.233	0.010	0.036	-0.124
24. Duration	0.106	0.127	-0.028	-0.213	0.130
25. Year	-0.081	-0.043	0.631	0.444	0.085
26. Age	-0.024	0.096	-0.007	0.120	-0.058

	11	12	13	14	15
11. Innovation	1.000				
12. Risk-taking	0.607	1.000			
13. Scanning	0.250	0.020	1.000		
14. Analysis	0.213	-0.157	0.610	1.000	
15. Multiplexity	0.270	-0.152	0.653	0.761	1.000
16. Integration	0.128	-0.261	0.606	0.487	-0.600
17. Futurity	0.618	0.302	0.359	0.462	0.477
18. Explicitness	0.186	-0.033	0.321	0.244	0.319
19. Adaptiveness	0.347	-0.016	0.672	0.602	0.736
20. Traditions	-0.442	-0.458	-0.213	-0.248	-0.216
21. Past success	0.192	0.147	-0.104	-0.092	-0.080
22. Current success	0.215	0.029	0.294	0.150	0.299
23. Succession	-0.122	-0.092	-0.026	0.111	-0.095
24. Duration	0.060	-0.049	0.168	0.144	0.140
25. Date	0.246	0.212	0.172	0.220	0.161
26. Age	-0.097	-0.102	0.302	0.214	-0.208
	16	17	18	19	20
16. Integration	1.000				
17. Futurity	0.268	1.000			
18. Explicitness	0.522	0.388	1.000		
19. Adaptiveness	0.605	0.401	0.380	1.000	
20. Traditions	0.085	-0.382	0.132	-0.196	1.000
21. Past success	-0.089	0.031	0.158	-0.160	0.127
22. Current success	0.330	0.030	0.247	0.382	0.035
23. Succession	-0.115	-0.163	-0.239	-0.129	-0.059
24. Duration	0.169	-0.053	0.031	0.203	0.137
25. Date	0.071	0.219	-0.020	0.066	-0.237
26. Age	0.184	0.011	0.043	0.129	0.089
	21	22	23	24	25
21. Past success	1.000				
22. Current success	0.113	1.000			
23. Succession	0.032	0.176	1.000		
24. Duration	-0.004	-0.178	-0.029	1.000	
25. Date	-0.021	0.053	-0.107	-0.115	1.000
26. Age	-0.036	0.113	-0.107	0.262	0.135

APPENDIX 3: PARTIAL LISTING OF DATA SOURCES

Below is a very incomplete list of the data sources used. It excludes the great many articles from the *New York Times*, *Wall Street Journal*, *Business Week*, *Forbes*, *Barron's*, *Printer's Ink*, *Advertising Age*, the *Economist*, and various trade publications, and financial data from Value Line, and Standard & Poor's.

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- Boise Cascade Corp.: Robert Katz, *Cases and Concepts in Corporate Strategy*. Englewood Cliffs, N.J.: Prentice-Hall, 1970; *Fortune*, Vols. 72 and 80.
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- Caterpillar Tractor: D. Buchanan, *Organizations in the Computer Age*. Aldershot: Gower, 1983; Annual reports, *Fortune*, vols. 17 and 85.
- Consolidated Edison: A. Lurkis, *The Power Bank*, New York: Icare Press, 1982; Annual reports, *Fortune*, vols. 73 and 90.
- Control Data Corp: Annual reports, *Fortune*, vols. 73 and 77.
- E. I. DuPont: G. Zilg, *Behind the Nylon Curtain*. Englewood Cliffs: Prentice-Hall, 1974; A. D. Chandler, *Strategy & Structure*. Cambridge, Mass.: MIT Press, 1962; *Fortune*, vols. 42, 76, 87.
- Ford Motor Co.: A. Nevins and F. Hill, *Ford: Expansion and Challenge: 1915-1933*. New York: Scribners, 1957; A. Nevins and F. Hill, *Ford: Decline and Rebirth: 1933-1962*. New York: Scribners, 1963; D. Halberstam, *The Reckoning*. New York: Avon, 1986.
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- International Paper Co.: IPC, *International Paper Company: 50 Years After, 1898-1948*. New York: 1948; Annual reports, *Fortune*, vols. 16, 79.
- International Telephone and Telegraph Corp.: A. Sampson, *The Sovereign State of ITT*. New York: Stein & Day, 1973; H. Geneen, *Managing*. Garden City, N.Y.: Doubleday, 1984.
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- RCA Corp. Annual reports, *Fortune*, vols. 6, 38, 56, 66, 86.
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- Unilever Ltd.: C. Wilson, *The History of Unilever* (3 volumes). London: Cassell, 1954, 1968.
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- Waltham Watch Company: C. Moore, *Timing A Century: The History of the Waltham Watch Co.*, Cambridge, Mass.: Harvard University Press, 1945.
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- Whitin Machine Works: T. Navin, *The Whitin Machine Works Since 1831*, Cambridge, Mass.: Harvard University Press, 1950.
- Xerox Corp.: J. Dessauer, *My Years with Xerox*. New York: Doubleday, 1971; D. K. Smith, *Fumbling the Future*. New York: W. Morrow, 1988; Annual reports, *Fortune*, vols. 66, 74, 90.
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APPENDIX 4: DEFINITIONS OF THE VARIABLES

Environment

1. *Uncertainty* or dynamism in the environment is manifested by the amount and unpredictability of change in customer tastes, production or service technologies, and modes of competition in the firm's principal industries.
2. *Hostility* in the environment is evidenced by price, product, technological and distribution competition, severe regulatory restrictions, shortages of labour or raw materials, and unfavourable demographic trends (e.g. shrinking markets).
3. *Heterogeneity* in the environment is evidenced by differences in competitive tactics, customers tastes, product lines, channels of distribution, etc. across the firm's respective markets. These differences are only significant to the extent that they require different marketing, production or administrative practices.

Structure

4. *Controls* monitor the internal trends and incidents relevant to organizational performance. M.I.S., employee performance appraisals, quality controls, cost and profit centres, budgeting and cost accounting are types of control devices. Score high if there is much emphasis on such controls.
5. *Internal communications* reflect the openness and fidelity of information channels in the organization. A high score is given when information reaches decision-makers quickly, when it is relevant and undistorted, and when communication flows readily in top-down, bottom-up, and lateral directions.
6. *Centralization of strategy-making power* involves the distribution of power for making strategic decisions regarding acquisitions, diversification, major new product introductions, long-term goals, etc. Centralization is high if the top executive alone makes most of the decisions with a minimum of consultation, low if middle managers determine strategies by the default or intent of top executives (general manager and up).
7. *Delegation of operating authority* concerns the amount of authority transferred to lower and middle levels of management (any parties below vice president) for administration of the day-to-day operation of the business. Operating decisions involve equipment replacement, production planning, adjusting prices of goods, inventory purchases, hiring of lower-level personnel, etc.
8. *Technocratization* is evidenced by the number of highly trained staff specialists and professionally qualified people (accountants, engineers, scientists, doctors) as a percentage of number of employees.
9. *Organizational differentiation* measures the degree of difference among organizational divisions or departments in terms of their overall goals, marketing and production methods, and decision-making styles. The more disparate the divisions, the higher the score. Even functionally organized firms with only one division may have high levels of differentiation if there exist many different approaches to marketing, production and other activities within different departments.

Strategy-making Process

10. *Proactiveness of decisions* is determined by whether or not a firm shapes its environment by introducing new products, technologies, administrative techniques, etc. A reactive firm follows the leader while a proactive firm is the first to act.
11. *Product market innovation*. Does the firm seem particularly innovative in terms of the number or novelty of new products and services introduced and new markets entered?
12. *Risk-taking*. Is there evidence that top managers are risk averse (score low), or does the firm frequently make large and risky resource commitments, commitments that have a reasonable chance of costly failure?
13. *Scanning* involves the search for problems and opportunities in the external environment of the firm. Firms are to be scored in terms of the amount of scanning performed on consumer tastes, competition, technological and administrative developments, etc. Scanning may be done by staff departments, executives, the sales force, etc. The greater the number of factors tracked and the more widespread the participation in scanning activity, the higher the score.
14. *Analysis of major decisions*. Do decision-makers devote much reflective thought and deliberation to a problem and the array of proposed responses? Time spent on correlating symptoms to get at the root cause of problems and effort spent on generating solutions (good or bad) are examples of analysis. A low score is given when there is a rapid

and intuitive response to an issue (this response may be ideal or the worst possible). Evidence of analysis includes time delays, frequent meetings and discussions, the use of staff specialists, and the writing of lengthy reports.

15. *Multiplexity of decisions.* Do top managers address a broad or narrow range of factors in making strategic decisions? For example, in deciding whether to acquire a company, a multiplex strategist would consider marketing, financial, production, demographic, administrative, and other problems, whereas low multiplexity would be evidenced by a focus, say, on marketing factors alone.

16. *Integration of decisions.* Are actions in one area of the firm complementary or supportive of those in other areas, or are they conflicting and mutually inhibiting? High integration would result in (or from) a concerted and well co-ordinated strategy while low integration might be manifested by fragmented or clashing tactics (e.g. acquiring new companies when there is inadequate ability to finance or run them, or selling products that compete against each other).

17. *Futurity of decisions* concerns the extent to which the firm looks into the future in planning its strategies and operations. A distant time horizon (5 years) warrants a high score. A focus on crisis decision-making warrants a low score.

18. *Explicitness of strategies* concerns the degree of conscious commitment to an explicit corporate strategy (i.e. a set of objectives coupled with a number of stated means for attaining these). A low score is evidenced by unclear goals and muddling though. This variable was not measured in the questionnaires.

19. *Adaptiveness of decisions* reflects the responsiveness and appropriateness of decisions to external environmental conditions. For example, an adaptive pricing decision would take into account competitive strategies, customer buying habits, government regulations, etc. A nonadaptive decision (score low) would neglect an important set of external factors.

20. *Traditions.* Does the firm often reconsider its strategies (low score) or are these tied largely to precedent?

APPENDIX 5: IMPACT OF PERIOD DURATION ON THE FINDINGS

The following table summarizes the significance levels of the key interaction terms or other coefficients used to test our hypotheses. All analyses were run exactly as those reported in tables I to IV-2, with identical control variables. The only difference is the composition of the sample, which now selects for different period durations.

Hypothesis 1: Interaction terms for relative inertia: intervals following success *vs.* mediocrity (see table I).

Sample N	Full sample 112	6 yrs or less 75	1 to 5 yrs 63	Over 5 yrs 49
Control				
Communication	+	*	+	
Centralization				
Delegation	***	**	*	0.11
Technocrat'n		0.13	+	
Different'n				
Proactiveness	**	**	*	0.12
Innovation				
Risk-taking	***	***	***	*

Scanning	*	*	*	*
Analysis	0.10		*	
Multiplexity	***	**	**	*
Integration	**	*	*	*
Futurity	**	**	**	
Explicitness	***	**	***	0.13
Adaptiveness				
Traditions			0.10	

Hypothesis 2: Effect of past performance on extremes: Equally supported in all samples except the last (over 5 years), where significance level was 0.10.

Hypothesis 3: Decline in information processing: Equally and strongly supported in all samples, except the last, where findings were slightly weaker.

Hypothesis 4-1 and 4-2: Interaction of change in environment and past performance on structure and strategy-making process.

<i>Interaction terms</i>				
<i>Sample N</i>	<i>Full sample 112</i>	<i>6 yrs or less 75</i>	<i>1 to 5 yrs 63</i>	<i>Over 5 yrs 49</i>
Uncertainty X past performance, with:				
Communication		0.11	0.13	
Centralization	*	+	0.14	*
Delegation				
Technocrat'n		0.13	+	
Different'n	*	***	*	
Proactiveness	**	*	+	+
Innovation	*	*	**	
Risk-taking	*	**	**	
Scanning				
Multiplexity		0.14	0.12	
Traditions	*	**	*	
Hostility X past performance, with:				
Controls	*	+		
Communications	**	*	+	0.13
Technocrat'n	*	**	**	
Scanning				
Multiplexity	*	+		
Adaptiveness	+	0.14	+	
Traditions	*	*	+	
Heterogeneity X past performance, with:				
Centralization	*	+		
Differentiation	***	***	***	

+, *, **, and *** indicate levels of significance of 0.10, 0.05, 0.01 and 0.001 in predicted direction (numbers are non-significant coefficients near the 0.10 level).

NOTES

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- [1] Although Meyer and Zucker (1989) argue that poor performance will result in a stalemate under which self-interest paralyses the organization, this need not always be the case. Conflict challenges basic assumptions about goals and values, and it can motivate creative compromise. Also, when the parties in conflict are not equipotent or totally at odds, change is indeed possible. Moderation and compromise, rather than mere stagnation, can be a product of conflict (Miles and Cameron, 1982).

Meyer and Zucker (1989) call attention to the fact that even poorly performing organizations can endure for a long time and resist attempts to make them more efficient or to shut them down. They argue that poor performance can create conflict between the dominant coalition of owners or managers who might wish to change the company and 'dependent actors' such as workers, suppliers, etc. who want it to survive. The result is sustained low performance. They also argue that when the interests of all parties are aligned, then the organization tends to pursue performance maximization (p. 154). For many of the reasons given above, we believe that this pursuit, however earnest, will usually be temporary.

- [2] Researchers such as Tolbert and Zucker (1983) assess institutionalization by the extent to which an organization is unresponsive to its environment.
- [3] In addition, successful firms have the slack resources to buffer themselves from some of the uncertainties in their environments. They can use strategies such as vertical integration, mergers, and the co-operation of powerful parties to reduce their day-to-day responsiveness to externalities (Pfeffer and Salancik, 1978).
- [4] A simple correlation analysis showed that the lagged correlations for the group of past stars vastly exceed those of the former poor-to-mediocre ('average') performers for all 17 structure and process variables. Thirteen of these differences are significant at the 0.05 level or better.
- [5] Hypothesis 2 does not apply to any of the information processing variables, which, according to Hypothesis 3 are expected to decline – to head towards a low but not a high extreme.

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